Web Programming

Lab Assignment 09

By: Avyakth Shriram (24BCE651)

LAB 9 – HTML Code:

```
<!DOCTYPE html>
<html>
    <head>
        <title>Objects in JavaScript</title>
    </head>
    <body style="background-color:blueviolet;">
        <div class="container" style="padding:20vh 4vw;">
            <div class="inner-cont" style="text-align:center; padding:20px; border-
radius:2rem; background-color:pink; height:500px; width:90vw;">
                <h1>Welcome, <span id="name" style="color:aqua"></span></h1>
                <h1>Three Different Ways to Create an Object: </h1>
                <div class="obj-crean" style="display:flex; gap:50px; padding:20px;">
                    <div class="method1">
                        <h1>Object Literal Method</h1>
                        <form>
                            Id  : <input type="text" id="id1"/>
                            Name: <input type="text" id="name1"/>
                        </form>
                        <button onclick="obj1()" style="background-color:yellowgreen; margin-
top:20px; border-radius:5px; order:none; width:100px; height:35px;">Click me!</button>
                        <p id="para1"></p>
                    </div>
                    <div class="method2">
                        <h1>Create Instance Of Object</h1>
                        <form>
                            Id  : <input type="text" id="id2" name="id2"/>
                            Name: <input type="text" id="name2" name="name2"/>
                        </form>
                        <button style="background-color:yellowgreen; margin-top:20px; border-
radius:5px; border:none; width:100px; height:35px;"
onclick="obj2(document.getElementById('id2').value,document.getElementById('name2').value)">c
lick me</button>
                        <p id="para2"></p>
                    </div>
                    <div class="method3">
                        <h1>Object Constructor</h1>
                        <form>
                            Id  : <input type="text" id="id3"/>
                            Name: <input type="text" id="name3"/>
                        </form>
                        <button style="background-color:yellowgreen; margin-top:20px; border-
radius:5px; border:none; width:100px; height:35px;" onclick="obj3()">click me!</button>
```

```
                        <p id="para3"></p>
                    </div>
                </div>
            </div>
            <h1 style="color:white; text-align:center; margin-top:100px;">Use of JS
objects</h1>
            <div style="display:flex; gap:10%; justify-content:center; margin-top:30px;
margin-bottom:30px;">
                <div class="par2" style="text-align:center;">
                    <button style="background-color:yellowgreen; margin-top:20px; border-
radius:5px; border:none; width:100px; height:35px;" onclick="calculate()">Calculate</button>
                    <div id="props" style="background-color:pink; border-radius: 3rem;
padding:10px; width:300px; margin:auto; margin-top:20px;">
                        <p id="length">Length: </p><br/>
                        <p id="reverse">Reverse: </p><br/>
                        <p id="conc">Concatenation: </p><br/>
                        <p id="subs">Substring: </p><br/>
                        <p id="includes">Includes: </p><br/>
                        <p id="copy">Copy: </p><br/>
                        <p id="casecon">CaseConversion: </p><br/>
                    </div>
                </div>
                <div class="par2" style="text-align:center;">
                    <button style="background-color:yellowgreen; margin-top:20px; border-
radius:5px; border:none; width:100px; height:35px;"
onclick="calculateMath()">Calculate</button>
                    <div id="props" style="background-color:pink; border-radius: 3rem;
padding:10px; width:300px; margin:auto; margin-top:20px;">
                        <p id="math-pi">Math.PI: </p><br/>
                        <p id="math-round">Math.Round: </p><br/>
                        <p id="math-floor">Math.Floor: </p><br/>
                        <p id="math-ceil">Math.Ceil: </p><br/>
                        <p id="math-sqrt">Math.Sqrt: </p><br/>
                        <p id="math-pow">Math.Pow: </p><br/>
                    </div>
                </div>
                <div class="par2" style="text-align:center;">
                    <button style="background-color:yellowgreen; margin-top:20px; border-
radius:5px; border:none; width:100px; height:35px;"
onclick="calculateArray()">Calculate</button>
                    <div id="props" style="background-color:pink;border-radius:
3rem;padding:10px;width:300px;margin:auto;margin-top:20px;">
                        <p id="array-length">Array Length: </p><br/>
                        <p id="array-reverse">Reversed Array: </p><br/>
                        <p id="array-sorted">Sorted Array: </p><br/>
                        <p id="array-sum">Sum of Array: </p><br/>
                        <p id="array-min">Min Value: </p><br/>
                        <p id="array-max">Max Value: </p><br/>
                        <p id ="ith element">Ith Element: </p><br/>
                    </div>
```

```
                </div>
            </div>
            <div style="display:flex; margin:auto; gap:10%; color:white; justify-
content:center; margin-top:30px; margin-bottom:30px;">
                <div class="par2" style="text-align:center;">
                    <h1>Form Validation:</h1>
                    <div id="props" style="background-color:pink; border-radius: 3rem;
padding:10px; width:300px; margin:auto; margin-top:20px;">
                        <form>
                            User Name: <input type="text" id="userName"><br/><br/>
                            Age: <input type="text" id="age"><br/><br/>
                            Password: <input type="password" id="password" maxlength="10"
minlength="8" ><br/><br/>
                            Confirm Password: <input type ="password" id="confirm"
maxlength="10" minlength="8"><br/><br/>
                            Address: <textarea rows="2" cols="30"
id="Address"></textarea><br/><br/>
                            <input style="background-color:yellowgreen; margin-top:20px;
border-radius:5px; border:none; width:100px; height:35px;" type="button" onclick="sub()"
value="Submit here">
                            <input style="background-color:yellowgreen; margin-top:20px;
border-radius:5px; border:none; width:100px; height:35px;" type="reset" value="reset">
                        </form>
                    </div>
            </div>
            <div class="par2" style="text-align:center;">
                    <h1>Login page using form: </h1>
                    <div id="props" style="background-color:pink; border-radius:3rem;
padding:10px; width:300px; margin:auto; margin-top:20px;">
                        <form name="loginForm" onSubmit="return validateLoginForm()"
method="post">
                            <label for="username">Username: </label><br>
                            <input type="text" id="username" name="username"><br><br>

                            <label for="password">Password: </label><br>
                            <input type="password" id="password" name="password"><br><br>

                            <input style="background-color:yellowgreen; margin-top:20px;
border-radius:5px; border:none; width:100px; height:35px;" type="submit" value="Login">
                        </form>
                    </div>
            </div>
            <div class="par2" style="text-align:center;">
                    <h1>Form Using Regex: </h1>
                    <div id="props" style="background-color:pink; border-radius:3rem;
padding:10px; width:300px; margin:auto; margin-top:20px;">
                        <form name="loginForm" onSubmit="return validateLoginForm()"
method="post">
                            <label for="lettersField">Letters Field: </label><br>
```

```
                            <input type="text" id="lettersField" name="lettersField"
onkeypress="onlyLetters(event)"><br><br>
                        </form>
                        <h1 style="background-color:yellowgreen; margin-top:20px;
margin:auto; border-radius:5px; border:none; width:100px; height:25px; font-
size:15px;" >:Letters Only:</h1>
                    </div>
            </div>
            </div>
        </div>
        <script type="text/javascript" src="LAB9 - 24BCE5461.js"></script>
    </body>
</html>
```

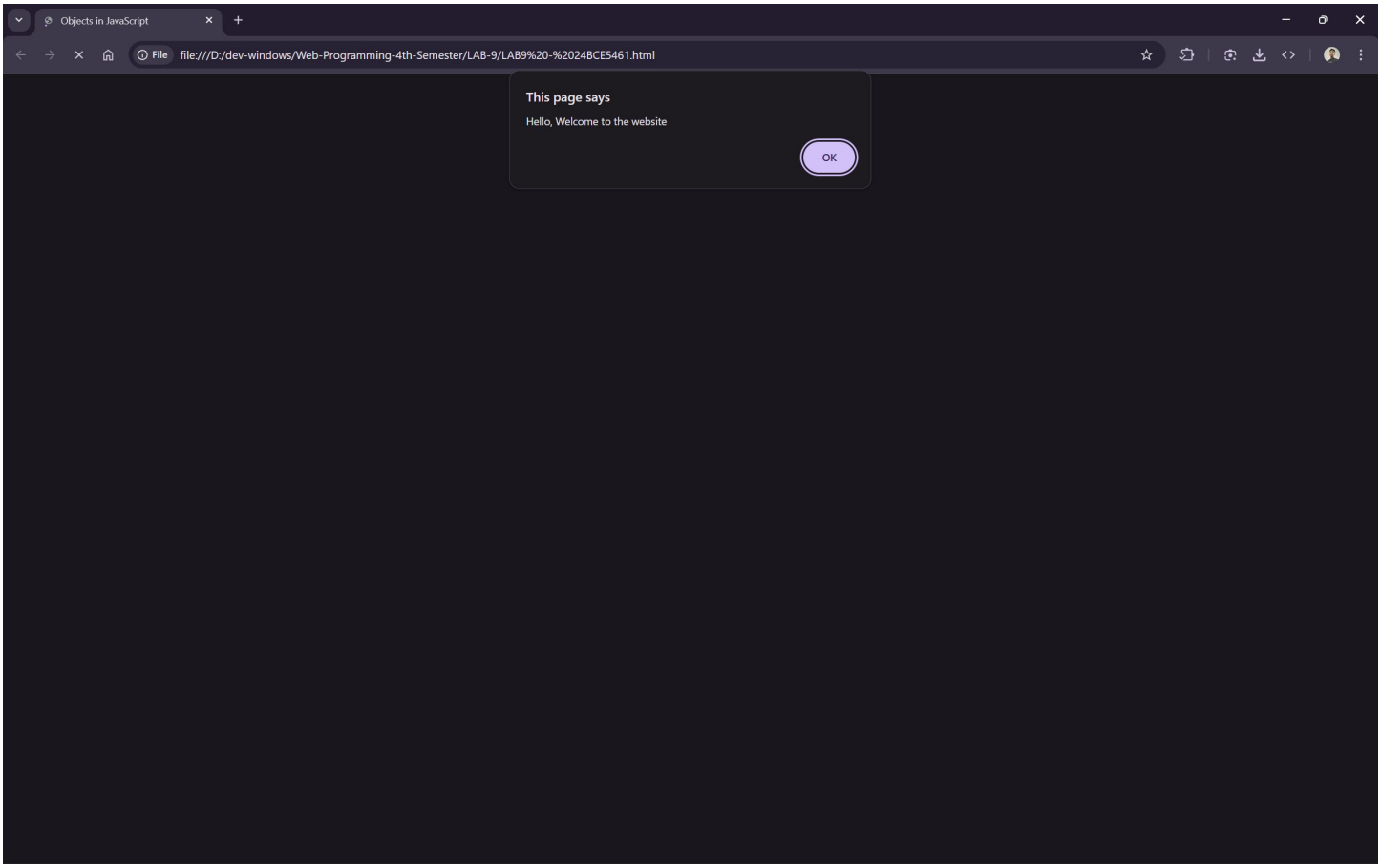
LAB 9 – JavaScript Code:

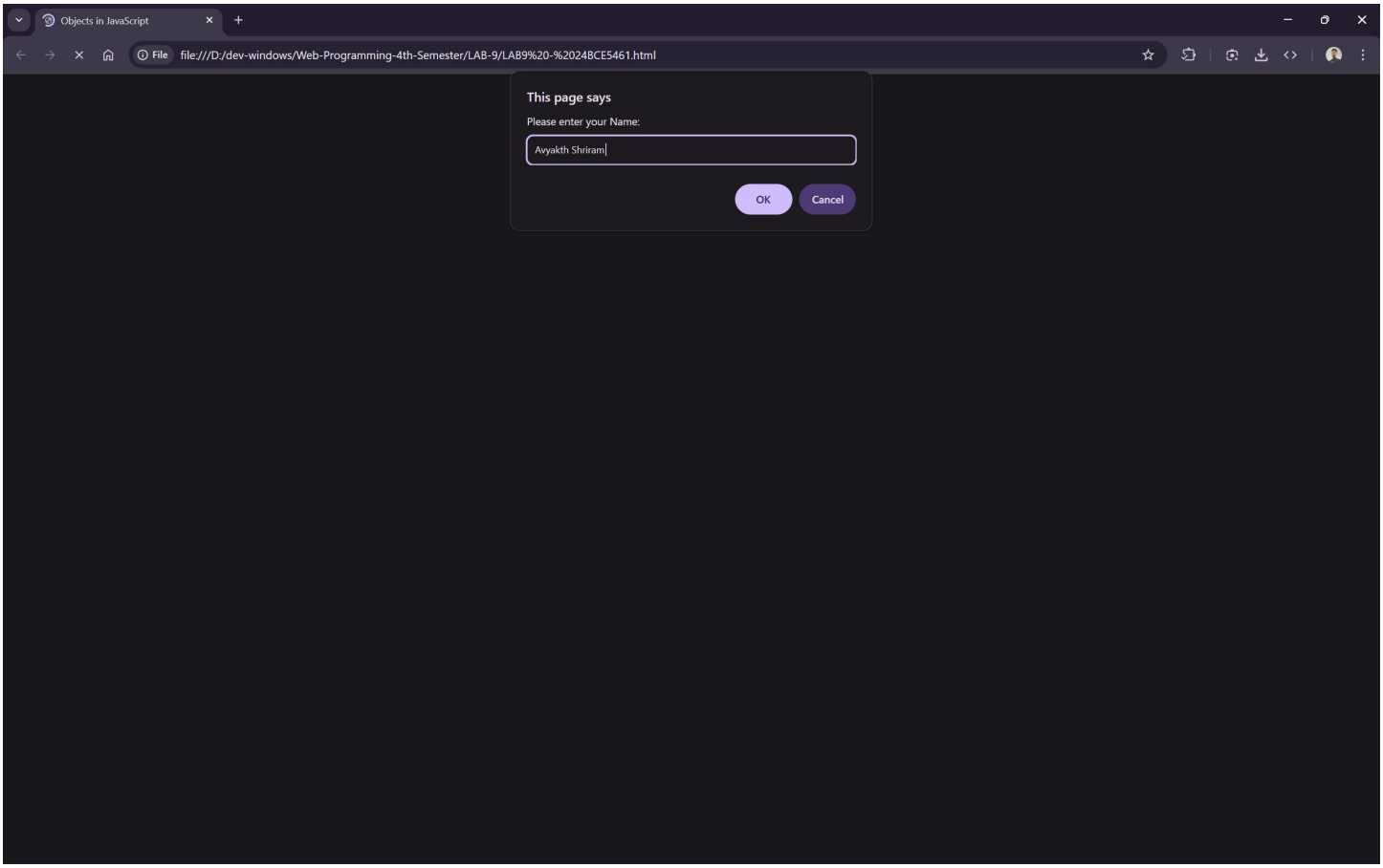
```
window.alert("Hello, Welcome to the website");
var name = prompt("Please enter your Name:")
document.getElementById("name").innerHTML = name;

function obj1() {
    var id1 = document.getElementById("id1").value ;
    var name1 = document.getElementById("name1").value;
    emp = {id:id1,name:name1}
    document.getElementById("para1").innerHTML = "empId: "+emp.id+" empName: "+emp.name;
}

function obj2(id2,name2) {
    var emp2 = new Object();
    emp2.id = id2;
    emp2.name = name2;
    document.getElementById("para2").innerHTML = "empId: "+emp2.id+" empName: "+emp2.name;
}

function obj3(){
    var id3 = document.getElementById("id3").value ;
    var name3 = document.getElementById("name3").value;
    function obj3(id,name) {
        this.id = id;
        this.name = name;
    }
    e = new obj3(id3,name3);
    document.getElementById("para3").innerHTML = "empId: "+e.id+" empName: "+e.name;
}

function calculate() {
    var str = prompt("Enter a string: ");
    var concatStr = prompt("Enter a string to concatenate: ");
    var substring_st_idx= parseInt(prompt("Enter the starting index for substring: "));
    var substring_end_idx= parseInt(prompt("Enter the ending index for substring: "));
    var includesChar = prompt("Enter a character to check if it is included in the string:
");
    document.getElementById("length").innerHTML = "Length: " + str.length;
    document.getElementById("reverse").innerHTML = "Reverse: " +
str.split("").reverse().join("");
    document.getElementById("conc").innerHTML = "Concatenation: " + str.concat(concatStr);
    document.getElementById("subs").innerHTML = "Substring: " +
str.substring(substring_st_idx, substring_end_idx);
```

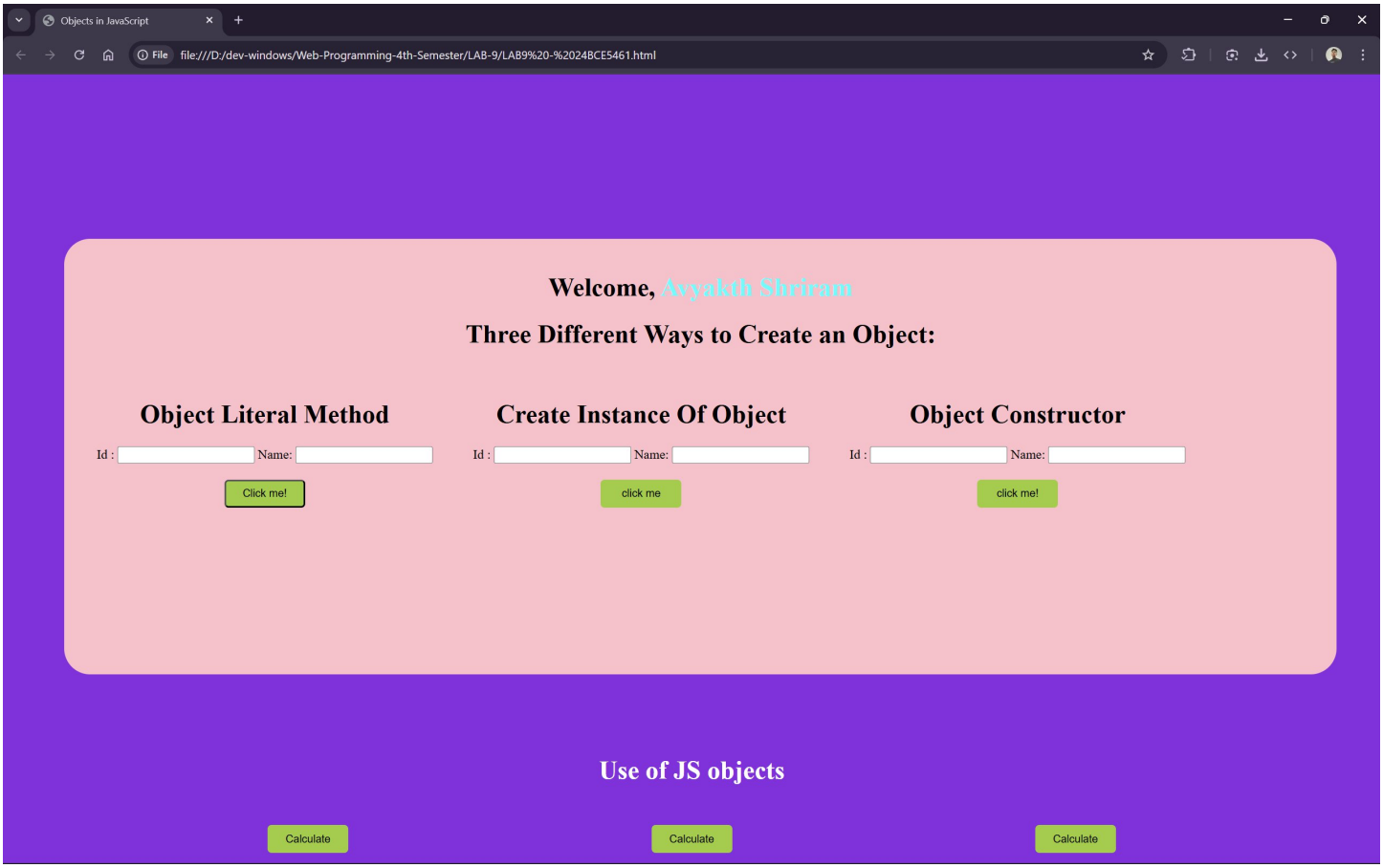
```
    document.getElementById("includes").innerHTML = "Includes '" + includesChar + "': " +
str.includes(includesChar);
    document.getElementById("copy").innerHTML = "Copy: " + str.slice(0);
    document.getElementById("casecon").innerHTML = "Case Conversion: " + str.toUpperCase();
}

function calculateMath() {
    var num = parseFloat(prompt("Enter a number: "));
    document.getElementById("math-pi").innerHTML = "Math.PI: " + Math.PI;
    document.getElementById("math-sqrt").innerHTML = "Square Root: " + Math.sqrt(num);
    document.getElementById("math-pow").innerHTML = "Power (num^2): " + Math.pow(num, 2);
    document.getElementById("math-round").innerHTML = "Rounded Value: " + Math.round(num);
    document.getElementById("math-floor").innerHTML = "Floor Value: " + Math.floor(num);
    document.getElementById("math-ceil").innerHTML = "Ceiling Value: " + Math.ceil(num);
}

function calculateArray() {
    var arr = prompt("Enter an array of numbers (comma separated): ").split(",").map(Number);
    document.getElementById("array-length").innerHTML = "Array Length: " + arr.length;
    document.getElementById("array-reverse").innerHTML = "Reversed Array: " +
arr.reverse().join(", ");
    document.getElementById("array-sorted").innerHTML = "Sorted Array: " + arr.sort((a, b) =>
a - b).join(", ");
    document.getElementById("array-sum").innerHTML = "Sum of Array: " + arr.reduce((sum, num)
=> sum + num, 0);
    document.getElementById("array-max").innerHTML = "Max Value: " + Math.max(...arr);
    document.getElementById("array-min").innerHTML = "Min Value: " + Math.min(...arr);
    var i = parseInt(prompt("Enter the index of the element you want to retrieve: "));
    document.getElementById("ith element").innerHTML = "Ith Element: " + arr[i];
}

function sub() {
    if (document.getElementById("userName").value == "") {
        alert("Please enter your name");
    } else if (isNaN(document.getElementById("age").value) ||
document.getElementById("age").value == "") {
        alert("Please enter a valid age");
    } else if (document.getElementById("confirm").value !=
document.getElementById("password").value) {
        alert("Passwords do not match");
    } else if (document.getElementById("Address").value == "") {
        alert("Please enter your address");
    } else {
        alert("Form submitted successfully!");
    }
}

function validateLoginForm() {
    var username = document.forms["loginForm"]["username"].value;
    var password = document.forms["loginForm"]["password"].value;
```

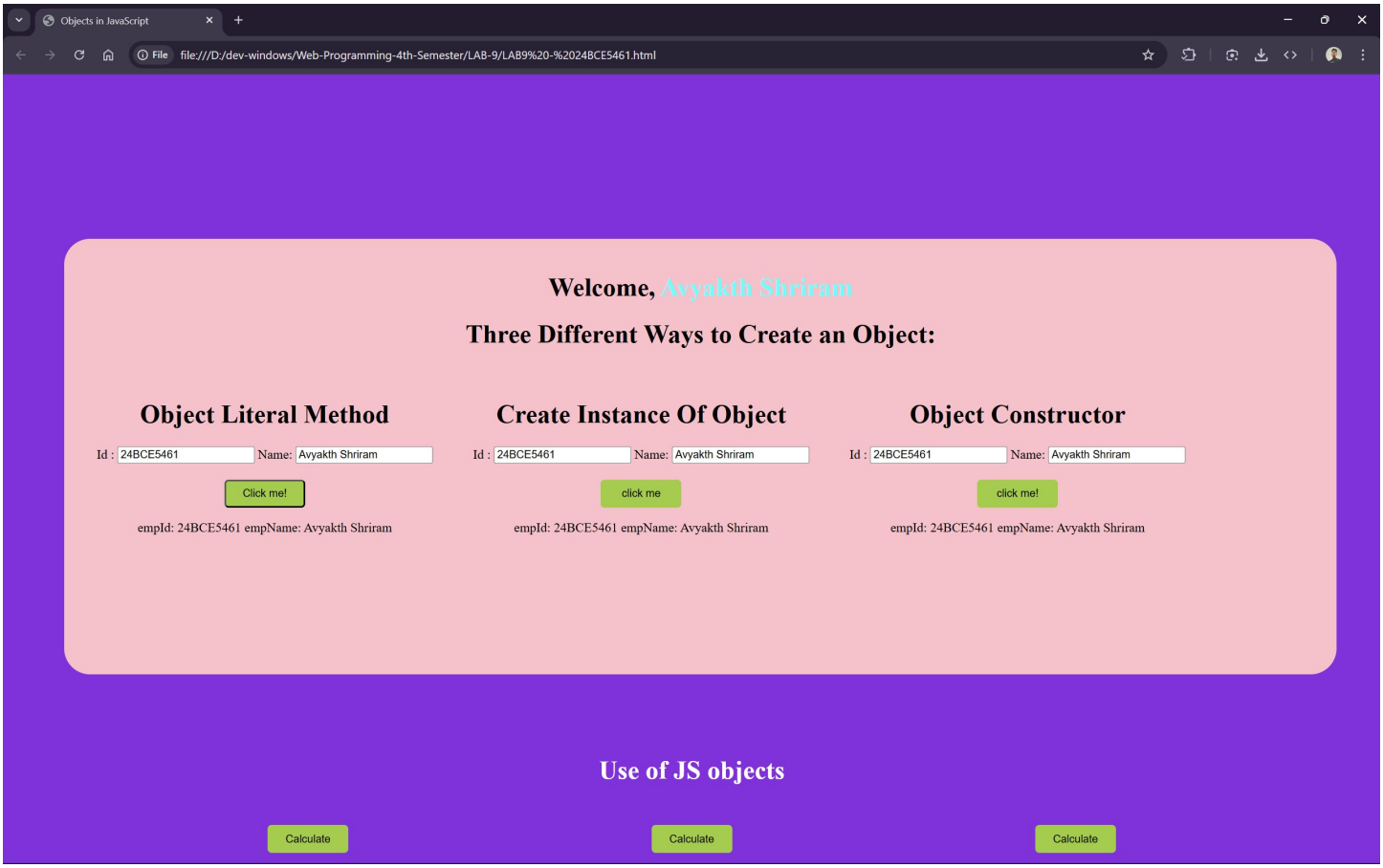
```
    if (username == "" || password == "") {
        alert("Please fill in all fields");
        return false;
    }
    if (password.length < 8) {
        alert("Password must be at least 8 characters long");
        return false;
    }
    alert("Login successful!");
    return true;
}

function onlyLetters(event) {
    var key = event.key;
    var regex = /[a-zA-Z]/;
    if (!regex.test(key)) {
        event.preventDefault();
        alert("Only letters are allowed");
    }
}
```

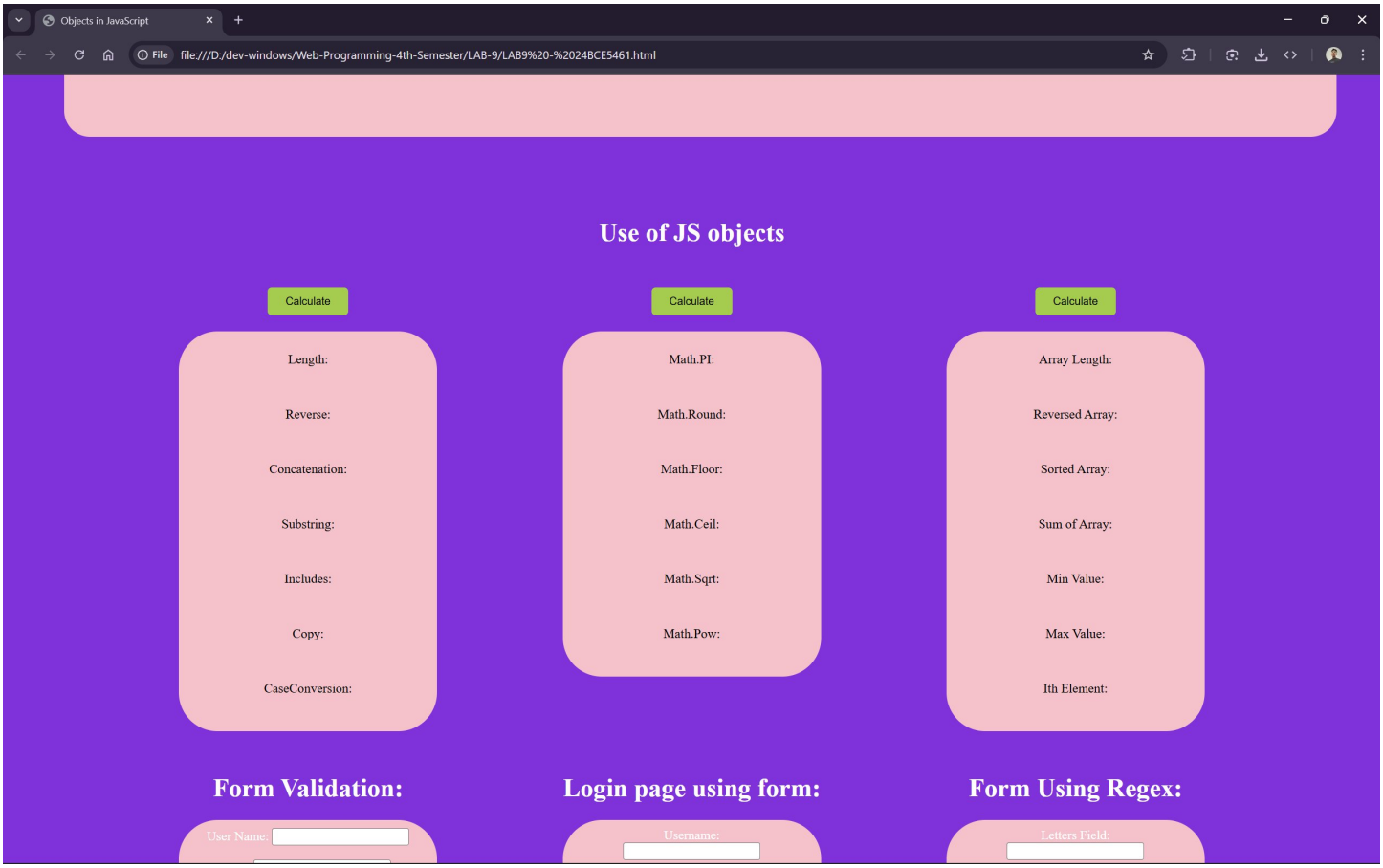
LAB 9 – Screenshots:

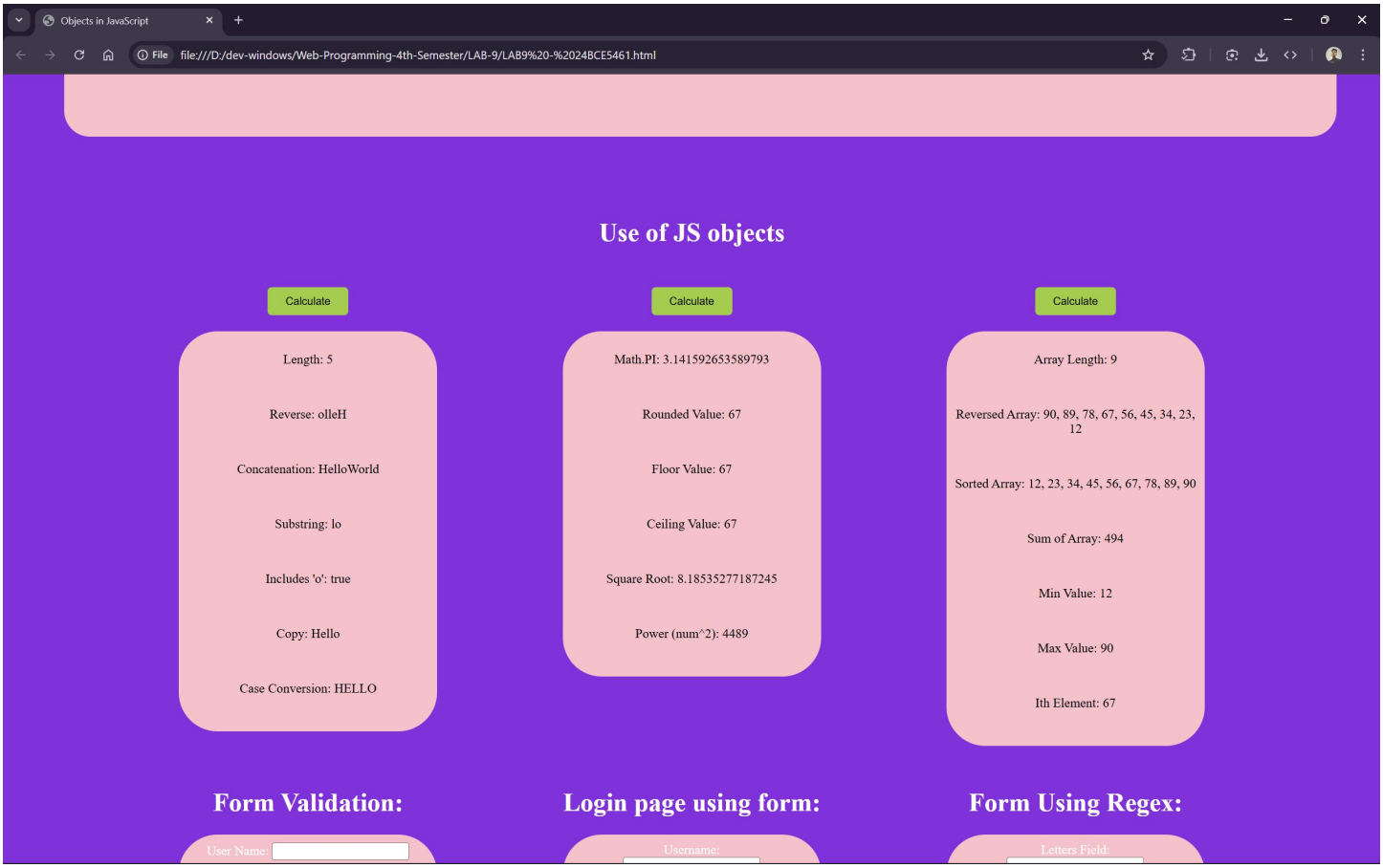


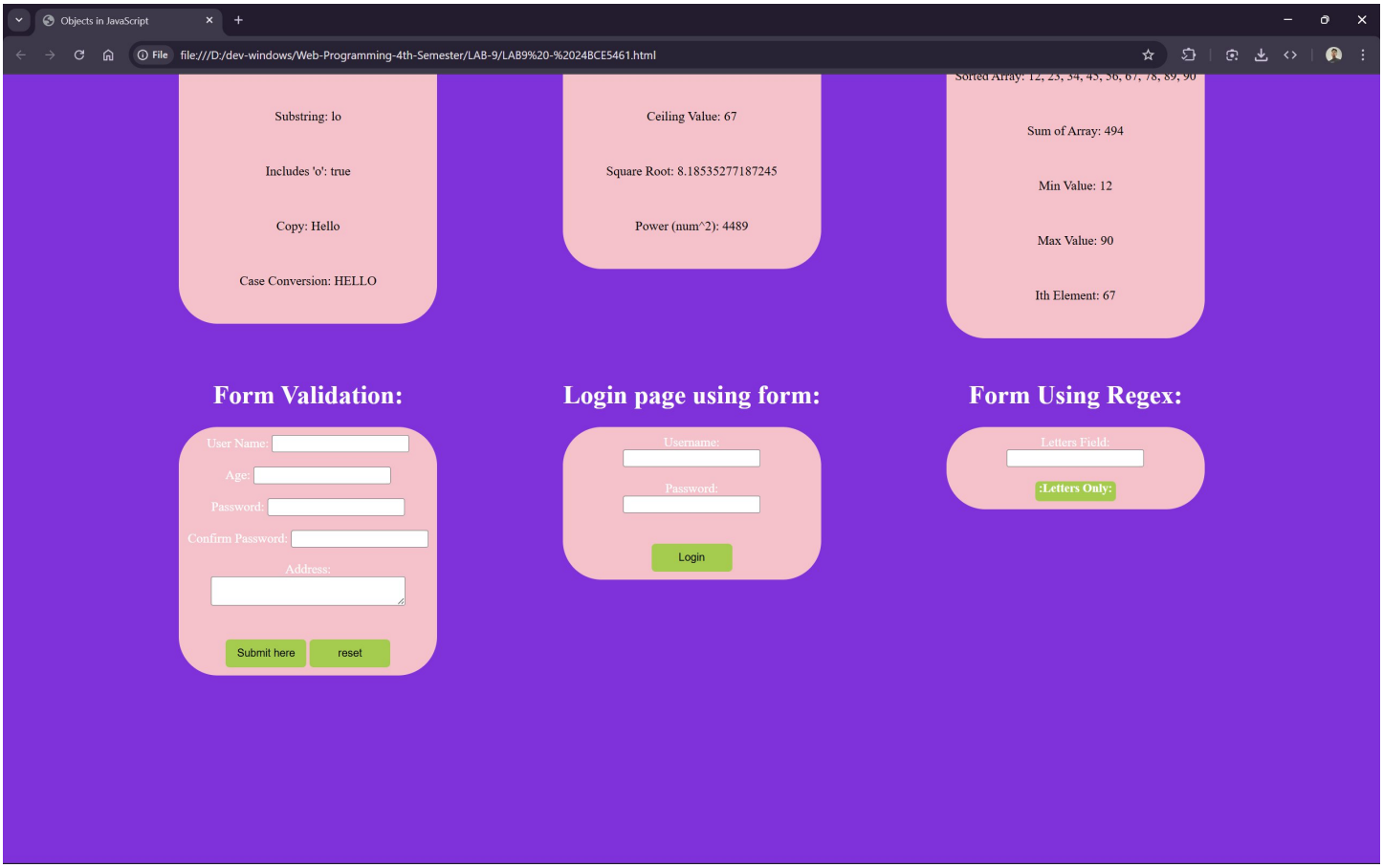

Objects in JavaScript
File
file:///D:/dev-windows/Web-Programming-4th-Semester/LAB-9/LAB9%20-%2024BCE5461.html
This page says
Please enter your Name:
Avyakth Shriram
OK
Cancel

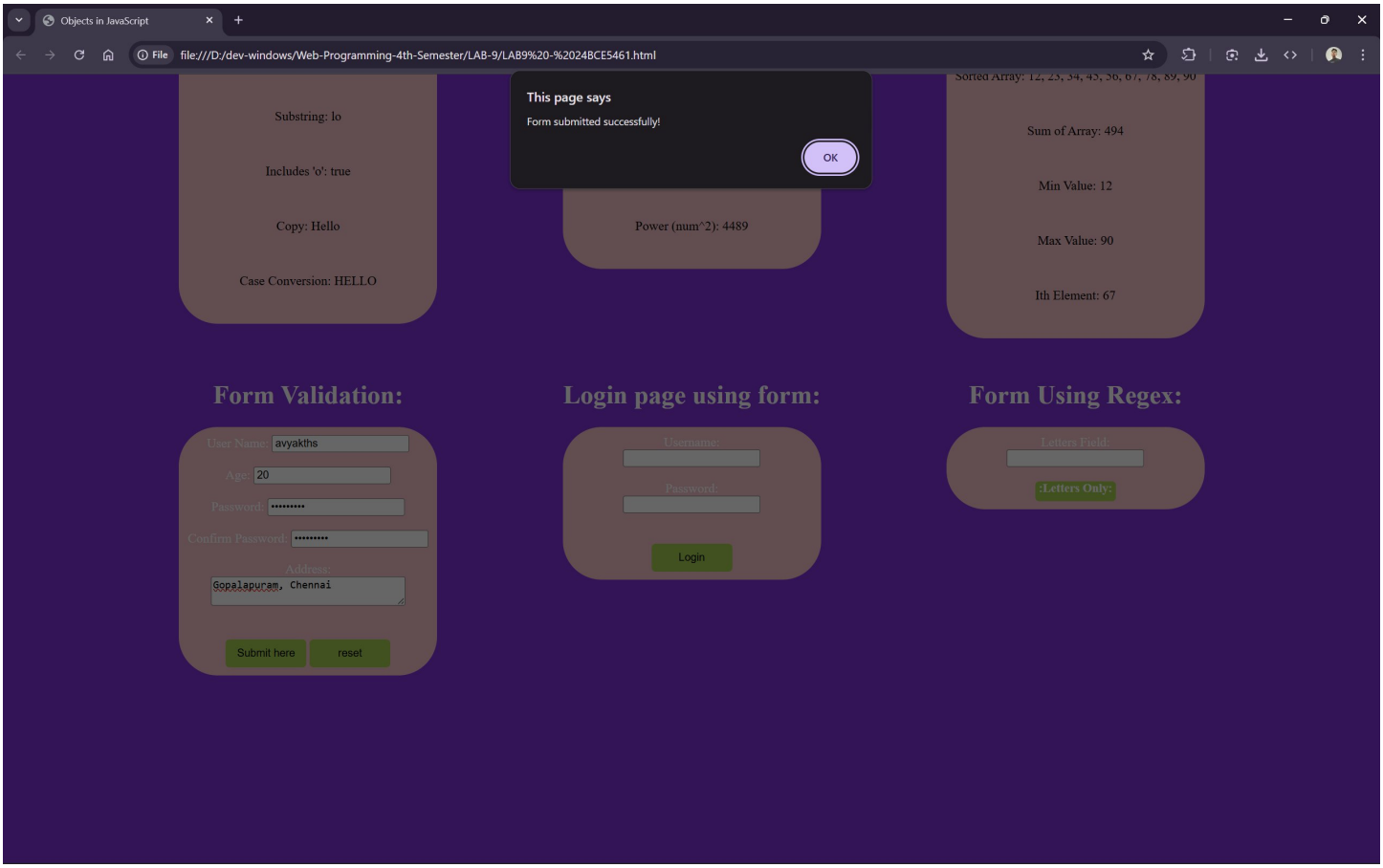

Objects in JavaScript
file:///D:/dev-windows/Web-Programming-4th-Semester/LAB-9/LAB9%20-%2024BCE5461.html
Welcome, Avyakth Shriram
Three Different Ways to Create an Object:
Object Literal Method
Id :
Name:
Click me!
Create Instance Of Object
Id :
Name:
click me
Object Constructor
Id :
Name:
click me!
Use of JS objects
Calculate
Calculate
Calculate

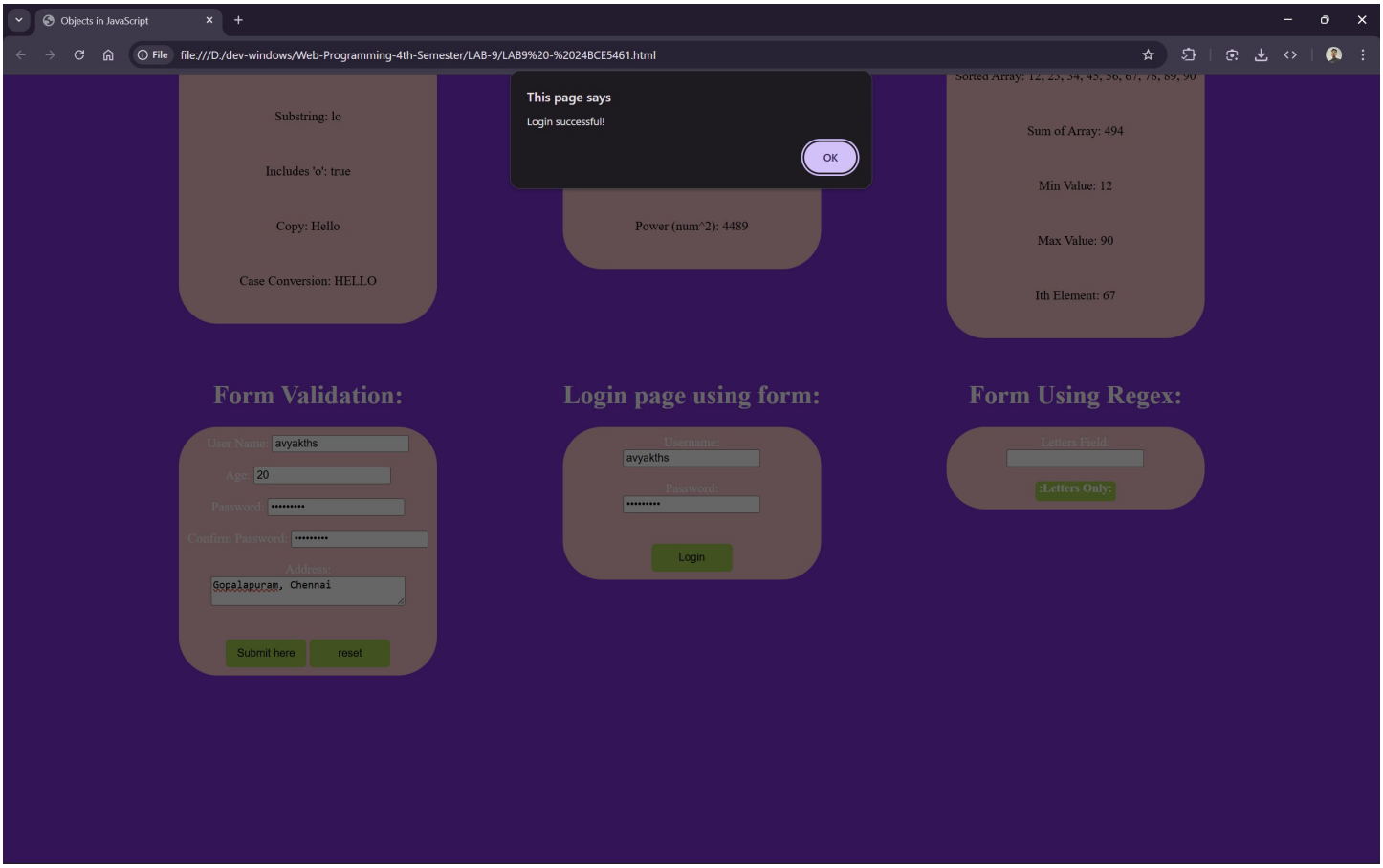

Objects in JavaScript
file:///D:/dev-windows/Web-Programming-4th-Semester/LAB-9/LAB9%20-%2024BCE5461.html
Welcome, Avyakth Shriram
Three Different Ways to Create an Object:
Object Literal Method
Id : 24BCE5461 Name: Avyakth Shriram
Click me!
empId: 24BCE5461 empName: Avyakth Shriram
Create Instance Of Object
Id : 24BCE5461 Name: Avyakth Shriram
click me
empId: 24BCE5461 empName: Avyakth Shriram
Object Constructor
Id : 24BCE5461 Name: Avyakth Shriram
click me!
empId: 24BCE5461 empName: Avyakth Shriram
Use of JS objects
Calculate
Calculate
Calculate


Objects in JavaScript
file:///D:/dev-windows/Web-Programming-4th-Semester/LAB-9/LAB9%20-%2024BCE5461.html
Use of JS objects
Calculate
Length:
Reverse:
Concatenation:
Substring:
Includes:
Copy:
CaseConversion:
Calculate
Math.PI:
Math.Round:
Math.Floor:
Math.Ceil:
Math.Sqrt:
Math.Pow:
Calculate
Array Length:
Reversed Array:
Sorted Array:
Sum of Array:
Min Value:
Max Value:
Ith Element:
Form Validation:
User Name:
Login page using form:
Username:
Form Using Regex:
Letters Field:


Objects in JavaScript
file:///D:/dev-windows/Web-Programming-4th-Semester/LAB-9/LAB9%20-%2024BCE5461.html
Use of JS objects
Calculate
Length: 5
Reverse: olleH
Concatenation: HelloWorld
Substring: lo
Includes 'o': true
Copy: Hello
Case Conversion: HELLO
Calculate
Math.PI: 3.141592653589793
Rounded Value: 67
Floor Value: 67
Ceiling Value: 67
Square Root: 8.18535277187245
Power (num^2): 4489
Calculate
Array Length: 9
Reversed Array: 90, 89, 78, 67, 56, 45, 34, 23, 12
Sorted Array: 12, 23, 34, 45, 56, 67, 78, 89, 90
Sum of Array: 494
Min Value: 12
Max Value: 90
Ith Element: 67
Form Validation:
User Name:
Login page using form:
Username:
Form Using Regex:
Letters Field:


Objects in JavaScript
file:///D:/dev-windows/Web-Programming-4th-Semester/LAB-9/LAB9%20-%2024BCE5461.html
Substring: lo
Includes 'o': true
Copy: Hello
Case Conversion: HELLO
Ceiling Value: 67
Square Root: 8.18535277187245
Power (num^2): 4489
Sum of Array: 494
Min Value: 12
Max Value: 90
Ith Element: 67
Form Validation:
User Name:
Age:
Password:
Confirm Password:
Address:
Submit here
reset
Login page using form:
Username:
Password:
Login
Form Using Regex:
Letters Field:
:Letters Only:


Objects in JavaScript
file:///D:/dev-windows/Web-Programming-4th-Semester/LAB-9/LAB9%20-%2024BCE5461.html
Substring: lo
Includes 'o': true
Copy: Hello
Case Conversion: HELLO
This page says
Form submitted successfully!
OK
Power (num^2): 4489
Sum of Array: 494
Min Value: 12
Max Value: 90
Ith Element: 67
Form Validation:
User Name:
avyakths
Age:
20
Password:
Confirm Password:
Address:
Gopalapuram, Chennai
Submit here
reset
Login page using form:
Username:
Password:
Login
Form Using Regex:
Letters Field:
:Letters Only:


Objects in JavaScript
file:///D:/dev-windows/Web-Programming-4th-Semester/LAB-9/LAB9%20-%2024BCE5461.html
Substring: lo
Includes 'o': true
Copy: Hello
Case Conversion: HELLO
This page says
Login successful!
OK
Power (num^2): 4489
Sum of Array: 494
Min Value: 12
Max Value: 90
Ith Element: 67
Form Validation:
User Name:
avyakths
Age:
20
Password:
Confirm Password:
Address:
Gopalapuram, Chennai
Submit here
reset
Login page using form:
Username:
avyakths
Password:
Login
Form Using Regex:
Letters Field:
:Letters Only:

THANK YOU!!!